

Designing Spiking Neural Networks with Spike Timing Plasticity Learning on Embedded System

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1. Summary of the Research

Background

Spiking Neural Networks (SNNs) with STDP learning offer energy-efficient, brain-inspired computation suitable for embedded systems.

My Research Goal

To develop an on-device SNN using unsupervised STDP for lightweight, low-power pattern and thermal image classification.

2. Approach/Methodology

2.1 Development

SNN with LIF neurons implemented in Python on Embedded Systems.

2.2 Requirements

Input : $M \times N$ tile patterns (Current: 3×3 . Target: 24×32 thermal images). Operates under severe memory limits.

2.3 Design (Specs)

Spikes via Bernoulli/Poisson.

Simplified STDP rule.

Array-based lightweight structure.

2.4 Evaluation

It classifies $M \times N$ patterns and performs on-device learning using STDP.



Firing Order	Time Difference (Δt)	Synaptic Change	Name
Pre spike → Post spike	$\Delta t > 0$	Whight is strengthened	LTP (Long-Term Potentiation)
Post spike → Pre spike	$\Delta t < 0$	Whight is weakened	LTD (Long-Term Depression)
No time window		no change	

3. Current Results and Status

Current Results:

An SNN was implemented from scratch on the Raspberry Pi Pico, currently operating on 3×3 pattern inputs. Its accuracy fluctuates between 40% and 70%, with improving stability as the main goal.

Indications:

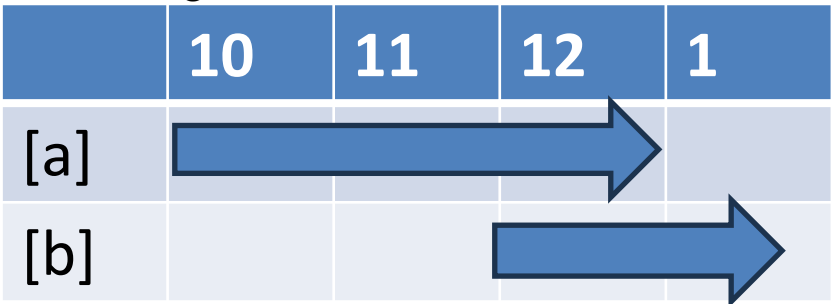
The feasibility of SNNs on microcontrollers was demonstrated, though inherent instability remains a key challenge.

Progress:

The study focuses on improving the accuracy and stability of the SNN and aims to deploy it for thermal image recognition.

4. Remaining Tasks and Tentative Schedule

1. Improve the accuracy of the $M \times N$ pattern classification
2. Extend the method to tiled thermal image classification.



[a]Extending to Thermal Image

[b]Thesis Preparation

References

- [1] **Diehl&Cook (2015)** Unsupervised learning of digit recognition using spike-timing-dependent plasticity